MENDSPEECH REVISED LEARNING SYSTEM

Week 3: Conformer From First Principles

Implement the core encoder pieces so model behavior is not a black box.

> Six focused build days plus one consolidation day. Each day ends with named artifacts or measured evidence.

Week milestone

> Implement the core encoder pieces so model behavior is not a black box.

Week map

Day	Focus	Minimum evidence	Compute
Day 15	Attention for speech sequences	• Change sequence length and measure forward time and memory.	Local CPU, L4 optional for scaling
Day 16	Conformer convolution module	• Feed synthetic impulses and inspect how local information spreads.	Local CPU
Day 17	Macaron feed forward and residual scaling	• Compare output statistics with and without residual scaling.	Local CPU
Day 18	Assemble one Conformer block	• Run forward and backward tests on several sequence lengths. • Intentionally remove one residual path and compare training stability on a toy task.	Local CPU
Day 19	Build a tiny Conformer encoder	• Track tensor shape through every layer on real speech. • Profile increasing depth.	Local CPU, L4 optional
Day 20	Compare your block with a production	• Choose one difference and reproduce its effect on a small benchmark if feasible.	Local CPU
Day 21	Week 3 architecture review	• Give yourself a ten minute whiteboard explanation from waveform features through one Conformer block.	Local CPU

Reference spine

- Conformer primary paper
- A mature Conformer implementation such as NVIDIA NeMo

DAY 15

Attention for speech sequences

Compute
Local CPU, L4 optional for scaling

LEARN

- Query, key, value projections.
- Scaled dot product attention.
- Attention masks.
- Sequence length cost.

BUILD IN MENDSPEECH

- Implement single head attention and then multi head attention in PyTorch.
- Add shape assertions and gradient tests.

EXPERIMENT AND MEASURE

- Change sequence length and measure forward time and memory.

REQUIRED OUTPUT

- src/models/attention.py
- tests/test_attention.py
- results/day15_attention_scaling.csv

> **Completion check:** You can derive every major tensor shape and explain quadratic sequence cost.

Study method
25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

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DAY 16

Conformer convolution module

Compute
Local CPU

LEARN

- Pointwise convolution.
- GLU gating.
- Depthwise convolution.
- Batch normalization and activation.
- Why local patterns matter in speech.

BUILD IN MENDSPEECH

- Implement a Conformer style convolution module.
- Test causality assumptions and receptive field.

EXPERIMENT AND MEASURE

- Feed synthetic impulses and inspect how local information spreads.

REQUIRED OUTPUT

- src/models/conformer_conv.py
- tests/test_conformer_conv.py
- notebooks/day16_receptive_field.ipynb

> **Completion check:** You can explain why depthwise convolution is computationally attractive and what local context it captures.

Study method
25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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DAY 17

Macaron feed forward and residual scaling

Compute
Local CPU

LEARN

- Feed forward expansion.
- Swish or SiLU activation.
- Dropout.
- Half step residual weighting in Conformer.

BUILD IN MENDSPEECH

- Implement the feed forward module and residual wrapper.
- Add numerical tests for shape and gradient flow.

EXPERIMENT AND MEASURE

- Compare output statistics with and without residual scaling.

REQUIRED OUTPUT

- src/models/conformer_ffn.py
- tests/test_conformer_ffn.py

> **Completion check:** You can explain the ordering of the Conformer block without memorizing a diagram.

Study method
25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

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DAY 18

Assemble one Conformer block

Compute
Local CPU

LEARN

- Macaron structure.
- Layer normalization placement.
- Attention plus convolution interaction.

BUILD IN MENDSPEECH

- Assemble feed forward, attention, convolution, second feed forward, and final normalization.
- Match expected input and output shapes.

EXPERIMENT AND MEASURE

- Run forward and backward tests on several sequence lengths.
- Intentionally remove one residual path and compare training stability on a toy task.

REQUIRED OUTPUT

- src/models/conformer_block.py
- tests/test_conformer_block.py
- docs/conformer_block_walkthrough.md

> **Completion check:** You can point to every operation and say why it exists.

Study method
25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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DAY 19

Build a tiny Conformer encoder

Compute
Local CPU, L4 optional

LEARN

- Input projection.
- Stacked blocks.
- Mask propagation.
- Temporal dimensions.

BUILD IN MENDSPEECH

- Build a small encoder around your blocks.
- Connect log Mel features to the encoder.

EXPERIMENT AND MEASURE

- Track tensor shape through every layer on real speech.
- Profile increasing depth.

REQUIRED OUTPUT

- src/models/tiny_conformer.py
- results/day19_shape_trace.md

Completion check: A real log Mel tensor can pass through your encoder and produce valid gradients.

Study method
25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

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DAY 20

Compare your block with a production

Compute
Local CPU

LEARN

- Read the original Conformer paper sections relevant to block design.
- Inspect a mature implementation such as NeMo.
- Identify differences caused by engineering and efficiency.

BUILD IN MENDSPEECH

- Create an annotated comparison table: your component, paper definition, production implementation.

EXPERIMENT AND MEASURE

- Choose one difference and reproduce its effect on a small benchmark if feasible.

REQUIRED OUTPUT

- docs/day20_implementation_comparison.md

> **Completion check:** You can read production Conformer code and orient yourself without treating it as magic.

Study method
25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

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DAY 21

Week 3 architecture review

Compute
Local CPU

LEARN

- Review attention, convolution, feed forward, normalization, residual paths, and sequence cost.

BUILD IN MENDSPEECH

- Add an architecture inspector page to MendSpeech showing encoder stage shapes and context assumptions.

EXPERIMENT AND MEASURE

- Give yourself a ten minute whiteboard explanation from waveform features through one Conformer block.

REQUIRED OUTPUT

- app/encoder_inspector.py
- reports/week3_conformer.md

> **Completion check:** You can explain which parts are local, which are global, and which become problematic for streaming.

Study method
25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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